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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

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1. Team & Project Overview

Team Name	Quintuplets
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 8 - Promote sustained, inclusive and sustainable economic growth
Project / Solution Name	Waitly
Name (Leader)	Dias Zhumadilda
Email & Phone (Leader)	cys2509080@xmu.edu.my
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

There are nearly 1.3 million micro, small and medium enterprises (MSMEs) in Malaysia.

This makes up 96% of all businesses and over 77% are micro enterprises. They employ 48.7% of the countries workforce.

[1][2] Most of these are handled by individuals like home cooks, small kitchens and caterers.

Orders are placed using their chat apps.

Questions about their menu (ada pedas tak?), quantity, pickup/delivery, are all answered by the owner manually.

Their address, total and payment are noted. Customers will type in Malay, English or Manglish.

As the owner is cooking most of the time mistakes and typos frequently occur. Totals aren't noted.

The owner has to match the transfer with the one in their banking app. These businesses

don't have any staff to do this. No payment gateway integration.

This is an important part of the economy as time is a key factor. The owner is spending time on this. Every minute they spend typing they aren't cooking/selling.

This means they are losing out on income and growth as they aren't able to reply in a minute.

Support is requested by SDG 8. Only 2/5 businesses are less digitised.

7 in 10 medium enterprises claim to be digitally successful [3]. General chatbots are risky.

A language model can generate menu items, prices and even fake that they have been paid for. Waitly solves this issue by having a cheap ordering assistant that knows localbut cannot remember facts and about money.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

The user can place orders from a small eatery by conversing with a Telegram bot. The owner needs to only verify the payment. The bot is shared across all shops, and each eatery has their own unique link and QR code. When the customer interacts with the bot, the bot will greet them with the shop's name and menu. The AI agent will answer any menu related queries (Qwen2.5-7B-Instruct). It can understand Manglish (e.g., "boss nak 2 nasi lemak tapau"). The agent will create an order based on the customer's query. Rules were set for the design. The model is not allowed to make any changes to the order except through using tools (add_item, set_order_type, request_payment etc). The FastAPI backend will validate all of these requests using a PostgreSQL database. The items must exist and be available, prices pulled from the menu, calculations done in Python and only changes to the order status done via code. There is a response guard in place to prevent any messages with tool names or mentions of payment being sent. Once the order is complete, the bot will display the shop's DuitNow QR code, including the amount and reference code. The owner will receive the order on their dashboard made using Next.js. They will check the bank and press either Approve or Not received. The customer will receive a standard confirmation message. If there are any complaints or requests for refunds, these will be passed onto the owner. The AI will not respond until Resume AI.

From the SDC Phase 1 workshops, the following were used: GenAI APIs (OpenAI compatible tool calling using our RunPod endpoint), database architecture (multi-tenancy PostgreSQL schema with Alembic migrations) and cloud deployment (API, webhook and dashboard deployed onto a Replit Reserved VM, inference done on RunPod).

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: Next.js 16 (React 19, TypeScript), Tailwind CSS 4, shadcn/ui, TanStack Query.

Owner dashboard (live orders, menu editing, settings).

Backend: Python 3.12, FastAPI, SQLModel, Alembic. JWT authentication using Argon2/bcrypt password hashing. Server Sent Events used to update live dashboard.

DB: PostgreSQL 16 (stores businesses, menus, customers, conversations, messages, orders, media files).

Chatbot: Telegram Bot API (webhook receives messages, bot replies and DuitNow QR images).

AI/Computing: Qwen2.5-7B-Instruct RunPod GPU endpoint. Called via the OpenAI compatible Chat Completions API with tool calling capability (openai Python SDK).

Deployment: Replit Reserved VM. Single process handles API requests, Telegram webhook, media and static dashboard exports. All keys are stored in Replit Secrets.

Architecture Diagram:

[Telegram App (Customer)] - [Telegram Bot API] - [Webhook] - [FastAPI Backend (Replit)] - [Agent Loop (Prompt + Tools)] - [RunPod Endpoint (Qwen2.5-7B-Instruct)]

[PostgreSQL DB (Menus, Orders, Chats)] - [Owner] - [Next.js Dashboard] - [FASTAPI Backend (REST + Live Events)]

[FASTAPI Backend] - [Telegram (QR and static confirmations)]

Official platforms provided by sponsors of SDC: Yes. Replit was used for deployment of the entire prototype. The Reserved VM provides a managed PostgreSQL database and Secrets storage. RunPod was used for the GPU endpoint. All customer turns are sent to the Qwen2.5-7B-Instruct model via the OpenAI compatible Chat API with tool schemas.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise & Define

We have designed our solution around a chat based food business process (Message > Price > Total > Address > Check Payment). The owner only requires to check the bank transfer and click approve.

Ideate

Since small food sellers are unable to incorporate a payment gateway, we decided to make the project a telegram bot where users manually verify their DuitNow payment. The AI cannot manipulate the data since we have only allowed it to call tools.

Prototype

We have built a prototype of this end to end system. There is a demo store called ‘Kak Ani Kitchen’. There is also an order, menu and setting dashboard. We have developed a development simulator to simulate the pipeline.

Testing

The project has over 100 unit tests to test the backend. We have made a script to simulate an LLM. We have also tested 8 scenarios for real conversations. We have noticed errors in the model such as it saying "dah bayar" which implies it has been paid for. We have added guards to the backend.

Deployment

The migration and seeding process is done by the Replit Reserved VM whenever the project starts. This means the webhook gets registered automatically. We have one URL for the dashboard and API so we can focus on getting the pilot stores and model quality done in phase 2.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 8.3: Implement policies to support productive activity, entrepreneurship, creativity and innovation to enhance formalisation and growth of micro-, small- and medium-sized enterprises. Target 8.2 (enabling): Increase economic productivity through technological upgrading and innovation [1] Bernama, "Govt Shifts Focus To Scaling Up MSMEs After Resilient 2025" (DOSM/SME Corp Data), 31 Jul 2026. Available at: [2] Business Today, "MSME Sector Outpaces Broader Economy In 2025…" (DOSM Data), 30 Jul 2026. Available at: [3] UOB Business Outlook Study 2026 (Malaysia) H1. Available at: [4] United Nations, Target 8 of Sustainable Development Goals.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?*	There are nearly 1.3 million MSMEs in Malaysia (96% of all businesses, more than 77% are micro). These provide employment to 48.7% of workers [1] [2].

Metric Category	Your Response
e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."	Initial situation: (to be verified in Phase 2) The team has estimated that all (100%) of the orders placed via the chat function are done manually by the owner (reading messages, providing quote, calculating total, ensuring payment). This takes approximately 5 minutes per order. Verification will be done by timing manual ordering process via at least 3 home food sellers.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Primary metrics: Owner processing time (minutes) per order completed Secondary metrics: % of orders with verified item, pricing and totals via code(aim is to have no prices generated by AI). % of tests passed using scripts
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Reduction in owner effort per order of ≥ 60% (about 5 minutes to ≤ 2 minutes) in the final prototype. 100% of order totals calculated by code and ≥ 90% of evaluation.conversations completed without owner inputting any reply.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

- [x] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [x] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Qwen2.5-7B-Instruct (Alibaba Cloud Qwen team)	**model**	**Apache-2.0**
FastAPI, Uvicorn, Starlette	**code (backend framework)**	**MIT / BSD-3-Clause**
SQLModel, Pydantic, Alembic	**code (ORM, validation, migrations)**	**MIT**
psycopg 3	**code (database driver)**	**LGPL-3.0**
OpenAI Python SDK	**code (LLM client)**	**Apache-2.0**

PyJWT, pwdlib (argon2-cffi, bcrypt), cryptography	**code (auth, hashing, encryption)**	**MIT / Apache-2.0 / BSD-3-Clause**
Next.js, React, TypeScript	**code (frontend)**	**MIT / Apache-2.0**
Tailwind CSS, shadcn/ui, Radix UI	**code (UI)**	**MIT**
TanStack Query, React Hook Form, Zod, sonner	**code (frontend libraries)**	**MIT**
lucide-react (icons)	**code**	**ISC**
PostgreSQL 16	**software (database)**	**PostgreSQL License**
Datasets	**dataset**	**None – all demo and test data is written by the team**

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team’s knowledge and that the project will be original work created during the SDC 2026 hackathon period.



25.10.2026